Gradient and the Equation of a Line

Name _____ Date _____ No calculator Version 1

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KEY WORDS gradient *(rōnaki)* · rise · run · steep · y-intercept *(pūtahi-y)* · equation of a line

A Rise over run ●○○

Remember: Gradient = rise ÷ run. A line that goes down from left to right has a negative gradient.

1. Write the gradient of each line. Simplify any fractions.

a) A line goes up 6 for every 2 across _____

b) A line goes up 4 for every 1 across _____

c) A line goes down 3 for every 1 across _____

d) A line goes up 1 for every 2 across _____

e) A line goes up 8 for every 4 across _____

f) A line goes down 10 for every 5 across _____

g) A line goes up 3 for every 4 across _____

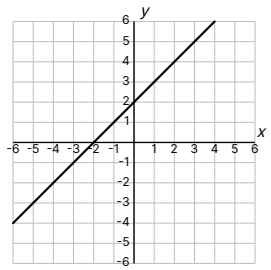
h) A line goes down 2 for every 6 across _____

B Gradient from a graph ●●○

Remember: Choose two points where the line crosses the corners of the grid squares. Count the rise and the run between them.

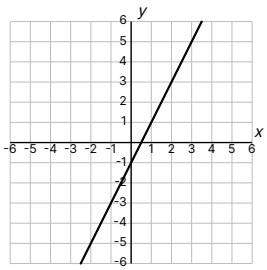
2. Find the gradient of each line.

a)



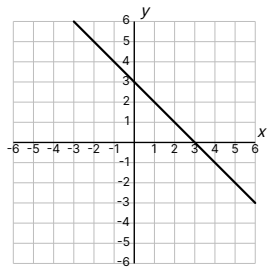
Gradient = _____

b)



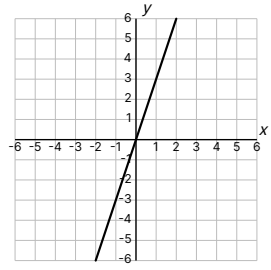
Gradient = _____

c)



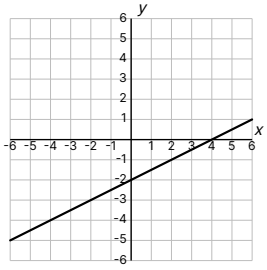
Gradient = _____

d)



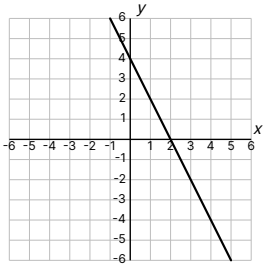
Gradient = _____

e)



Gradient = _____

f)



Gradient = _____

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C Gradient and y-intercept from a table ●●○

Remember: The gradient is how much y changes when x goes up by 1. The y-intercept is the value of y when $x = 0$.

3. Find the gradient and the y-intercept of the line for each table.

a)

x	0	1	2	3	4
y	1	3	5	7	9

b)

x	0	1	2	3	4
y	-2	1	4	7	10

c)

x	0	1	2	3	4
y	6	5	4	3	2

d)

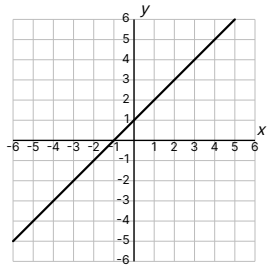
x	-1	0	1	2	3
y	-4	0	4	8	12

D The equation of a line from its graph ●●●

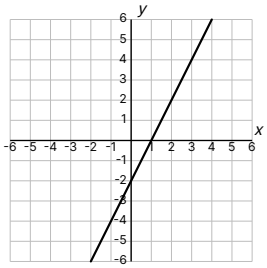
Remember: Find the gradient m and the y-intercept c, then write $y = mx + c$.

4. Write the equation of each line.

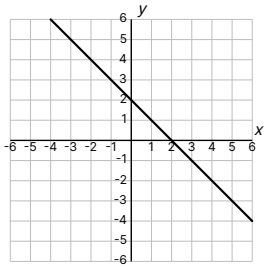
a)



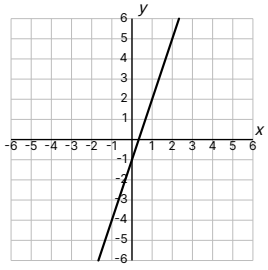
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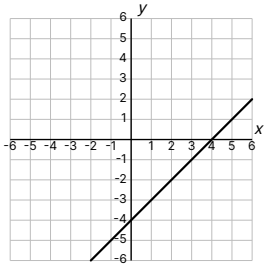
c)



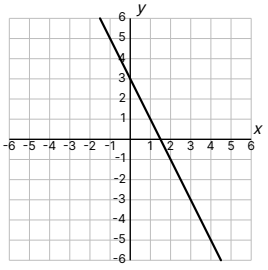
d)



e)



f)



★ Extension

5. These tables do not show $x = 0$. Find the equation of the line for each table.

a)

x	1	2	3	4	5
y	5	7	9	11	13

b)

x	2	3	4	5	6
y	2	5	8	11	14

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Gradient and the Equation of a Line

Name _____ Date _____ No calculator Version 2

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KEY WORDS gradient *(rōnaki)* · rise · run · steep · y-intercept *(pūtahi-y)* · equation of a line

A Rise over run ●○○

Remember: Gradient = rise ÷ run. A line that goes down from left to right has a negative gradient.

1. Write the gradient of each line. Simplify any fractions.

a) A line goes up 8 for every 2 across _____

b) A line goes up 5 for every 1 across _____

c) A line goes down 2 for every 1 across _____

d) A line goes up 1 for every 3 across _____

e) A line goes up 9 for every 3 across _____

f) A line goes down 12 for every 4 across _____

g) A line goes up 2 for every 5 across _____

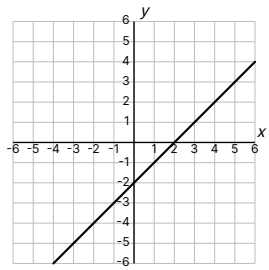
h) A line goes down 3 for every 6 across _____

B Gradient from a graph ●●○

Remember: Choose two points where the line crosses the corners of the grid squares. Count the rise and the run between them.

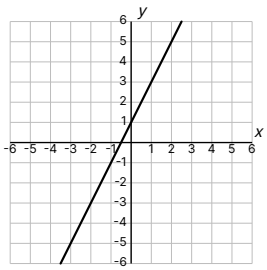
2. Find the gradient of each line.

a)



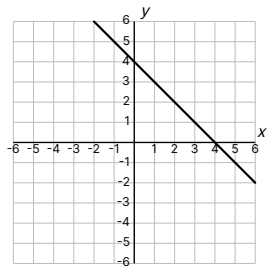
Gradient = _____

b)



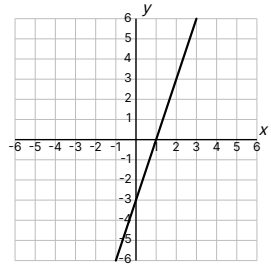
Gradient = _____

c)



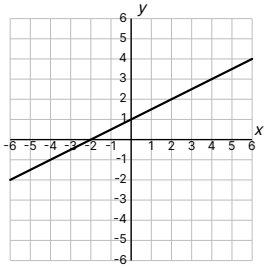
Gradient = _____

d)



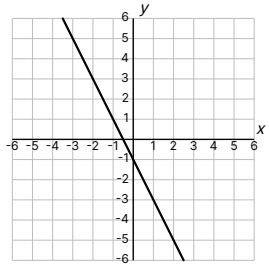
Gradient = _____

e)



Gradient = _____

f)



Gradient = _____

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C Gradient and y-intercept from a table ●●○

Remember: The gradient is how much y changes when x goes up by 1. The y-intercept is the value of y when $x = 0$.

3. Find the gradient and the y-intercept of the line for each table.

a)

x	0	1	2	3	4
y	1	4	7	10	13

b)

x	0	1	2	3	4
y	-3	-1	1	3	5

c)

x	0	1	2	3	4
y	9	7	5	3	1

d)

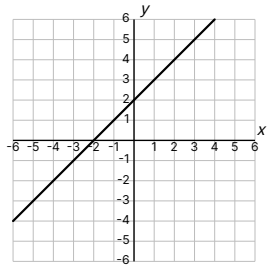
x	-1	0	1	2	3
y	-5	0	5	10	15

D The equation of a line from its graph ●●●

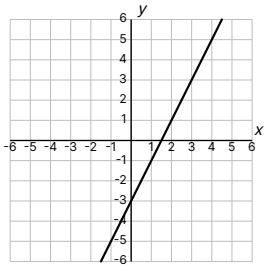
Remember: Find the gradient m and the y-intercept c, then write $y = mx + c$.

4. Write the equation of each line.

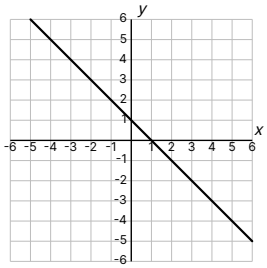
a)



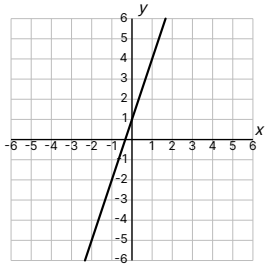
b)



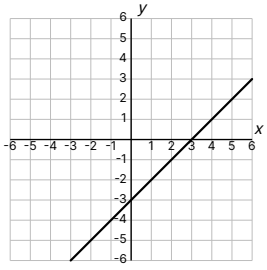
c)



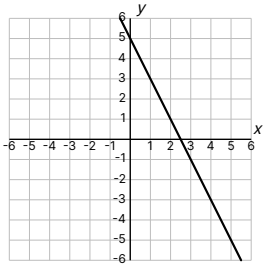
d)



e)



f)



★ Extension

5. These tables do not show $x = 0$. Find the equation of the line for each table.

a)

x	1	2	3	4	5
y	5	8	11	14	17

b)

x	2	3	4	5	6
y	-1	1	3	5	7

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